

Visionary.

A single-user studio for training LoRAs on your own photographs, generating stills with them, and animating those stills into clips — *one deploy, one URL, on your own Modal account.*

```
$ modal deploy app.py
```

STATUS	SURFACE	CATALOGUE	HARDWARE
Phases 1–4 complete	1 URL · 38 routes	22 weights · 4 families	H100 / H200

The core problem, and what this is worth

Three failures stack on top of each other. Only the third one is interesting.

The pipeline is three tools that do not know about each other

Training a LoRA on your photographs, generating stills with it, and animating one of those stills into a clip are, today, three installs — a trainer with its own dataset format, a node-graph UI with its own model directory, and a video stack with a third set of weights. Each has its own idea of where a caption lives and what a checkpoint is called. Moving between them is filesystem work, paid on every iteration, and a generated still that wants to become a clip goes out to disk and comes back by hand.

The GPU is rented for the wrong work

A persistent box sized for a 26 GB DiT idles at that size while it pulls files, writes thumbnails and validates a path. The obvious alternative — a hosted service — solves idling by taking the weights away from you: per-seat billing, your images on someone else's storage, and no way to get out a model you trained.

The unit of authorship is a string, and strings transfer

Prompt engineering is a programming skill wearing an artist's clothes. Because a prompt is copy-pasteable, somebody else's method transfers whole — take their string, get their results, generate endless variations in their voice without ever having their taste.

Every image on Civitai looks the same, and it is not because the models cannot do anything else. It is because the prompt is the method, methods get shared, and a shared method converges. That is what an aesthetic looks like when its unit of authorship can be pasted.

There is a quieter version of the same failure. Every model arrives with its own grammar — MiniMax-H3 wants a six-field document published in its own repo, Wan wants prose, Krea 2 wants prose with the camera clauses dropped. Asking a person to hold three formats in their head so a checkpoint can be fed correctly is the tool making its problem into theirs. A documented grammar presented as a blank textarea reads as superstition: whether a comma, or *the* woman versus *a* woman, changes the take is not something anyone can infer — and a video take is two to three minutes, so every guess is billed at that rate.

Why now

- **Open weights got good enough to own.** Krea 2 for stills, MiniMax-H3 for video that carries its own soundtrack in the same latent, Wan 2.2 for video that takes LoRAs. Not demos — checkpoints one person can train against and ship work from.

- **Serverless GPU made a single-user studio economically sane.** Per-second billing and scale-to-zero mean there is nothing to keep alive between sessions. The install is a deploy; the bill is the work.
- **Someone else solved regional LoRA, and solved it better.** Masking a single-stream DiT's attention used to need a vendored fork. A node pack now does it through ComfyUI's public hooks, and does it more strongly — multiplying each LoRA's activation delta by zero outside its box, so there is no pathway left for one character's identity to reach another's. The last reason this had to be a research project went away.

The value proposition

OWN THE WEIGHTS	Authorship at the prompt is thin — it is the last few centimetres of a pipeline somebody else built. Authorship in a LoRA trained on your own photographs is the model itself learning what you meant. That is why training is Phase 1 rather than an advanced feature bolted on later.
NO SETUP	No Modal Secrets, no CLI configuration, no database. The HuggingFace token is pasted into the UI. Nothing downloads on its own.
DATA STAYS A FOLDER	Datasets are images with <code>.txt</code> caption sidecars beside them — exactly what the trainer reads. No export step, no schema to fall out of sync with the files. Storage layout is the contract, not the code.
MODEL-AWARE UI	One shot vocabulary compiles to three grammars, and only the controls a chosen model actually reads appear on screen. A control that is present but ignored is worse than one that is absent — it is the interface making a promise the model will not keep.
COST DISCIPLINE	Downloads, uploads, thumbnails and validation run on CPU containers. A bad LoRA path is a form error in milliseconds rather than a cold start and 35 GB of weight loading before it fails.

Current feature architecture

Four phases complete and in this release. One Modal app, defined in one file.

How it is built — three words, in priority order when they conflict

ANTIFRAGILE	A failure should leave the system better able to survive the next one. Errors diagnose themselves: the missing-model error prints the resolved volume name, the exact path it wanted, and what is actually on the volume — the three facts that separate a wrong Modal profile from a filename typo from a partial download. Any error a user can hit twice should have explained itself the first time. Destructive operations are asked for rather than softened: deletion unlinks, and the dialog that says what is going and how much of it is the safety net. A per-root trash folder lived here for a while, until it was clear nothing ever surfaced it — it was not an undo anyone could reach, it was a second copy of everything already thrown away. Stops are cooperative. Pins are commit SHAs, so an upstream force-push cannot change your build under you.
SCALABLE	Not requests per second — dataset size, model size and cost per job. Never rent a GPU to do CPU work. Keep the polled thing small: a record polled every two seconds must never grow with the size of the result. Separate storage by commit cost. One container per loaded checkpoint.
FUTURE-PROOF	Prefer the surface that will still be there, and that carries the <i>next</i> model in for free. Maintained upstreams over owned forks. Storage layout is the contract, not the code. Separate images when pins conflict and only then — a dependency conflict resolved by compromise is one you pay for forever, and the converse binds just as hard, which is why images and video share an image rather than paying for a second CUDA build. Do not build a second way to do the first thing.

Runtime

SHAPE	One Modal app — UI, ~38 API routes and GPU jobs behind a single URL.
IMAGES	Three — trainer, inference/ComfyUI, captioner. Separate only because their <code>transformers</code> pins genuinely conflict; images and video share one image because nothing in it is per-family.
GPU CLASSES	Two, both Hopper — which is the bill for sharing one image, since SageAttention is compiled for sm_90. <code>max_containers=1</code> each, so a warm checkpoint is shared rather than reloaded.
VOLUMES	Two. The HuggingFace cache is separate because committing 17 GB of weights after writing 80 captions turned an 8-minute job into a 23-minute one.
UPSTREAMS	Two commit SHAs. Nothing vendored, nothing patched.

Storage — the volume is the contract

`models/` is flat and addressed by exact path. `loras/{folder}/` is what training writes, `datasets/` and `drafts/` are the same shape as each other, `outputs/{job}/` holds generated media, and the rest is disposable scratch.

A folder is a LoRA and the checkpoints in it are that LoRA's epochs — but anything arriving another way, migrated off an older volume or pulled off Drive, is a bare file at the top level, and a folders-only walk skipped four real LoRAs silently. That reads to the user as “training never produced anything” rather than “this listing has an opinion about directory layout.” A file a run can load is a file the picker has to offer.

Model catalogue and acquisition

Twenty-two entries across four families, every repo and filename verified against the HuggingFace API rather than copied out of documentation.

Krea 2 — images

RAW for training and Turbo for 8-step generation at 26.3 GB each, Qwen Image VAE, a bf16 Qwen3-VL 4B text encoder, and an optional identity-edit LoRA.

MiniMax-H3 — video with sound

Two int8 DiTs at 21 GB each (fl2va and ref2va), a 32B nvfp4 text encoder, a video VAE and an audio VAE.

Wan 2.2 — video

The A14B high/low-noise expert pair for t2v and i2v, the single 5B TI2V DiT, umT5-XXL, and both VAE generations.

Wan 2.2 speed LoRAs

Matched LightX2V 4-step high/low pairs, one set for t2v and one for i2v.

Acquisition is a first-class job rather than a script:

- **A family downloads itself.** The group is the unit you decide in; clicking files one at a time kept the queue in a person rather than in the program.
- **One download at a time, and being busy is a state rather than an error.** Three concurrent pulls measured 4–12 MB/s each against ~31 MB/s for one, so the route is idempotent: a second press returns the job the first one started.
- **243.8 MB/s, measured** — against 30.6 MB/s on the plain backend for the same 21 GB file. That 8× was the whole distance between a working day of downloads and two minutes, and it had been left on the table by an unset environment variable.
- **A long transfer publishes bytes.** A watcher polls bytes on disk and reports a count and a rate; a stall is abandoned and resumed. Without it, “stalled at 4 GB” and “running fine at 90 MB/s” were the same empty log.

- **Staged, then moved.** A half-written `.safetensors` is never visible to the picker, so it cannot be chosen and fail thirty seconds into a warm run.
- **A Google Drive route** on the same job contract — most LoRAs worth having were never published to HuggingFace. They are a link someone sent you.

Datasets and captioning

- **Saving is a choice.** Dropping images makes a draft that captions, filters and trains identically to a saved set. The page never asks for a name before the images are in front of you, because “is this worth keeping” is not a question you can answer at drop time. Drafts belong to the window that made them and are swept after fifteen minutes of silence, with the folder’s own mtime counted as a heartbeat so an upload still writing cannot be swept out from under itself.
- **Orientation is resolved on arrival.** An image carrying an EXIF rotation tag is rewritten upright once, so the page, the captioner and the trainer’s bucketer all see the same pixels. Left to each reader, this trains the rotation into the LoRA.
- **Five caption presets, each one rule inverted.** What a caption names is what the model learns is free to vary; what it never names is what the trigger word ends up owning. Character describes pose, wardrobe, framing and light and refuses to describe a face. Style describes the content and never the look. Concept describes the context around the thing.
- **A refusal is caught, not written.** A stock instruct model declines to describe photographs of real people often enough to matter, and a decline is fluent prose that passes every downstream check — so the symptom is a LoRA that learned to say it cannot describe someone. Refusals are detected before they reach a sidecar, counted by name, and the picker offers an ablated repack: same architecture, same loader, so the fix costs a repo id.

Training

Krea 2 LoRA training on musubi-tuner — rank, alpha, learning rate, epochs, save-every, resolution, flow shift and seed, defaulting to a rank-32 / alpha-32 / 1e-4 / 30-epoch run. Checkpoints land per epoch, so a folder is a LoRA and the files in it are its epochs. Progress rides on the Train door itself, because a run lasts hours and you are meant to leave and keep working.

Image generation

- Krea 2 Turbo at 8 steps and CFG 1.0 — the absence of a negative branch, not a low setting — and RAW at 28 steps and CFG 5.5, on a resident checkpoint.
- **LoRAs are written in the prompt**, in Automatic1111’s `<lora:name:0.8>` syntax, up to six. A name resolves by shortest unambiguous match, exact case first — two files differing only in capitalisation are two LoRAs, and folding case made both untypeable. A picker writes a token at the caret, because you cannot type a syntax you have never seen.

- **Regions are drawn on the canvas**, up to eight. Each box takes its own prompt, its own LoRA at its own strength, and optionally its own reference photograph — a latent mold that pulls that rectangle toward that face during sampling. A box with a photo and no LoRA is a character with no training run behind it.
- Scene and outfit transfer when the identity-edit LoRA is present.
- The ratio picker and the pixel boxes are one control, snapping to 8 on the way out because the pipeline floors to the VAE’s grid regardless — and a box that keeps 1000 while the model renders 992 is a box that lied to you.

Video — two families, one path

Adding Wan did not add a backend. It reuses the container, the warm ComfyUI process, the job contract and the output layout; what is per-family is a graph builder and a table row. They are not interchangeable, and the interface says so rather than averaging them.

	MINIMAX - H3	WAN 2.2
Audio	Native stereo, same latent	Silent
CFG / negative	No — guidance-distilled	Yes
LoRAs	No ecosystem for the int8 repackage	Yes
References	9 images, 12 across all types	No
Experts	One	Two on A14B, one on the 5B
Tiers	768p · 544p draft	720p / 704p · 480p draft
Length	5–14 s at 24 fps	2–5 s at 16 or 24 fps
Tasks	t2v, i2v, first/last, ref2va	t2v, i2v, first/last

The model table is served to the page, so the composer renders only what the chosen model reads. Two rules earned in this path are worth stating on their own:

- **Write out every optional input a node takes.** A ComfyUI node declares its defaults twice — once for the canvas widget, once in its `run()` signature — and for an input the graph omits, it is the signature that wins. In the pack we drive, two of them disagreed, so every plate render ran the identity-edit LoRA 43% hot and downscaled every reference to 1024. None of it was visible in the graph, the sidecar or the page. Spelling every optional input out turns a disagreement upstream can introduce silently into one a diff shows.

- **Every input a run is priced by needs a range, and a file does not come with one.** Tier, seconds, aspect and steps are all bounded. “Max detail” on a reference was not: the node floors the short edge at 2048 and caps nothing, so a 4032×3024 straight off a phone arrived as 2720×2048 — 21,760 latent tokens against the 3,996 the same file would have cost at “match” — and reference tokens ride *every* sampling step, so the run died in step 0 of 8. References are bounded on arrival at 1536px now. The general rule: an option whose cost is set by something the page never measured is an option that will be found by whoever has the biggest camera.

The prompt compiler

One vocabulary table with nine groups — framing, angle, light, tone, action, speech, camera, sound and score — and three compilers behind it. Each group declares which side reads it and where its clause lands. H3 gets the six-field document, with the alignment instruction held verbatim including the guide’s own inconsistencies, because it is a contract with the checkpoint rather than phrasing we chose. Wan gets prose with the audio pills dropped by capability rather than by field, since dialogue is the case that breaks the simpler rule. Krea 2 gets prose with camera, action and audio filtered — and the palette *dims* what the model cannot read, with the group heading saying why.

Pick nothing and the typed text passes through byte-for-byte. Every prompt written before this still means what it meant, and the document only appears once you have said something that needs one.

The compiler closes your sentence, chooses the separator in front of it, and does nothing else.

`non_diegetic_music: N/A` is the default, and is worth the feature on its own: H3 invented a soundtrack for every clip until something told it not to. The disclosure under the pill rail runs the same compiler on the same container, so it shows the exact string the encoder will be handed — a preview with its own implementation is worse than no preview. Reference chips carry a role, which is what makes “do not describe the picture you attached” enforceable rather than advice: there is somewhere else for that description to go, and it is one click.

The job contract, and what makes it survive

Every long operation — download, Drive pull, caption, train, generate, video — publishes to the same record and answers the same status and stop routes. New capability extends it rather than inventing a parallel one.

- **A job record has two writers, so publishing holds a lock.** The job thread writes phases; the log drain writes step counts. Interleaved without the lock, a stale read put `running` back over a `completed` in 15 modelled runs out of 15, and the page then polled a finished job forever.
- **A poll waits for its own reply.** The replacement for `setInterval`, which fires on a clock rather than on a reply — overlapping polls held connections against a six-per-origin budget and starved the very gallery covers and video byte-ranges they were painting.

- **A job record is a claim about a container, worth only what its age says.** Every publish stamps a beat, because a container killed mid-transfer never reaches its own terminal path — and the record outlives the deploy, so rebuilding the image cannot clear it.
- **A dead ComfyUI is diagnosed and replaced, never re-raised as a socket error.** A GPU fault reaches us as a connection reset. The liveness check runs before anything blames the transport, and a fresh process starts at the top of the next run rather than every render being refused until the scaledown window expires. An OOM triggers a reclaim charged to the job that already failed, rather than to every job to prevent one.
- **Stops are cooperative.** Jobs check a flag between steps and unwind cleanly, so the container survives and the next request is warm.

The interface

One page, no build step, no dependencies. The canvas is the largest thing on screen at every moment and the controls live in a bar under it — never a rail beside it, which costs the picture 384px of the one dimension it cannot get back. The bar is capped at 30% of the viewport, and the prompt field is what yields to the budget. Image and video are one workspace sharing prompt, canvas and gallery; the switch is a chip inside the prompt field and the sentence survives it. The gallery has three layers, its navigation lives in the top corners, and the last generation is a thumbnail beside Generate, where the hand that just pressed Generate already is.

The console pass has a measured result

Controls are sorted by how often you reach for them, not by whether they are per-take or per-session. Scope sounds right and fails on its own examples: nobody picks a new aspect ratio per render, and CFG is not something you set once a session either — it is something you almost never set. The seed is the case that proves it. It is different on every render, so by scope it belongs in the row; but nobody types a seed, you take it off a result. Rarely reached for, so behind the button, and no special pleading required.

CONSOLE STRIP	BEFORE	AFTER
Image	1016 px of controls	732 px
Video	979 px of controls	652 px

Controls also carry no chrome until the pointer is on them: the value is what makes a control a control, and every one of these already shows its own.

Bar versus rail is settled with evidence, not taste

Fitting each render aspect into the canvas at 1512×982 leaves **0px** of dead vertical space at all five aspects, and 152–1068px horizontal. The picture is height-bound everywhere, so the bar always comes out of the picture while four of five aspects leave room for a rail twice over. The bar stays anyway — a rail is a dated shape that will read as of-its-decade long before this app stops being useful, and the overlay that would have

avoided both was built and rejected, because a backdrop blur means judging a render through something whose legibility depends on what you generated. What the measurements really argued for was less in the bar, not the bar somewhere else.

Every picture the model can be given sits in one row

Keyframes and references are the same decision made two ways — they load different transformers, so one excludes the other. As two separate rows of unlabelled 36px tiles, the keyframe pair was never found at all, and dropping photos into the reference tray looked like filling keyframe slots that kept growing. Side by side with a rule between them, the tray that grows and the two slots that do not are told apart by shape, which is the thing a tooltip could not do. Whichever half is out of play goes dim rather than disappearing: a control that vanishes when you fill its neighbour teaches nothing except that the page lost it.

A local preview server runs the page against stubbed JSON shaped to hold the awkward states — a missing model, an uncaptioned dataset, a prompt too long to belong in a gallery card — so front-end work costs a reload rather than an image build and a cold start.

The long-term vision & roadmap

The console is scaffolding. It is what made the rest legible, and it is expected to go.

The end state is a canvas where you argue with the picture instead of authoring a prompt — and the prompt itself is an implementation detail nobody should ever have to see.

Horizon 1 – NEXT TWO QUARTERS

Phase 5 · video LoRA training

Wan 2.2 is the target, which is why the video path already loads LoRAs and why a video LoRA row already carries an expert. This extends the existing trainer rather than adding a second one, and it is the piece that makes “own the weights” true on the video side as well as the image side.

2K, when and only when it ships locally

2K is absent on purpose. H3-Regenerate-2K is not a super-resolution module — it feeds the 768p result *plus the original multimodal context* back into the base model, which is what recovers small text and fine detail that conventional upscaling has to guess. MiniMax withholds it. Building against their hosted endpoint was considered and rejected: it sends renders to a third party, bills outside Modal, and becomes dead code the day the local module lands. One thing to build first when it does — upscaling this app’s own output beats upscaling a dropped file, because the method’s whole advantage is the original context, and our sidecar still holds the prompt, the pills and the references.

Horizon 2 – PHASE 6, THE DYNAMIC CANVAS

The canvas becomes where the work is done rather than where the result appears. Every attachment is a drop: a photo onto a box is that character, a photo onto the frame is the scene, and keyframes and references arrive the same way instead of through a tray of 32px tiles. What you touch answers with its own affordances, at the place you touched it, and only the ones that object can respond to — not a menu at a fixed corner, which is a panel in miniature parked at a coordinate.

The capability is honestly tiered, and the tiers cost very differently:

TIER	WHAT IT IS	STATUS
Reachable now	Touch-to-select and touch-to-edit. Segmentation on a finished render turns flat pixels into addressable regions; regional inpainting re-renders one mask.	An engineering step. The regional path is already element-as-component in primitive form — the gap is that boxes are author-declared rather than model-derived.

TIER	WHAT IT IS	STATUS
Reachable, expensive	Elements that persist across renders.	A LoRA already gives this for a character. Making it general means a LoRA per element and a training run per thing kept — which changes what a library is: weights, not images.
Not there	A frame computed from a spatial arrangement rather than authored, and a camera with real physical limits.	Causality between a 3D scene and a 2D image, which diffusion does not model. Needs a world model, or a hybrid where real 3D drives conditioning. Nothing off the shelf does it.

The plan chases the first tier and leaves the third alone until the model exists. Attempting it, failing, and concluding the whole direction is fantasy is the specific mistake worth naming in advance.

Horizon 3 – THE END STATE, AND WHAT IT VETOES

These are design constants, and each one exists to kill a specific cheap fix. The cheap fix is always a panel. If a rule here has never vetoed anything, it is not earning its place.

- **If a gesture cannot communicate intent without a label, the answer is a better gesture.**
Kills the control panel — and the tutorial. Coach marks and first-run overlays are labels in a costume.
- **The canvas never changes.** Character, wardrobe, action, location, camera — one surface. What changes is what the surface is willing to hear.
Kills tabs, rooms, modes, a back button. Today's Generate/Train split is on the wrong side of this.
- **What you touch decides the mode.** You do not enter character mode, you touch a face.
Kills tool palettes, mode switchers, selected-tool state.
- **Everything is live, nothing is labelled.** Every element addressable at all times, none of them drawn as a control.
Kills inspectors, sidebars, property panels — which is what the console is.
- **Nothing asks for confirmation.** Move something and the frame changes.
Kills modals and the Generate button. The trade is that undo must be total, because reversibility is what replaces confirmation.
- **Duration starts at zero.** A still is the default and time is added.
Kills a timeline on first run. Someone who wants one image should finish and leave without learning motion exists.
- **The frame is computed, never authored.**
Kills composing by dragging contents around a viewport. You move the world, not the picture of it.

- **The camera is a character.** It has a want and real limits; it can be late, it can look away.
Kills the camera as a settings group and lens choice as taste.
- **Derived or invented, always visible.**
Kills the chat panel and clarifying questions. Every question asked is a small failure — pick something, mark it invented, move on.
- **The library is closed until you reach for it.** A drawer, not a workspace.
Kills the persistent rail, the docked browser, the asset manager.
- **It never remembers unless told.**
Kills personalisation and learned defaults. The first screen of the thousandth session is the first screen of the first.

The strategic bet underneath all of it

A prompt is a compilation target, not something the user writes. Half of this is already built and is the proof it works: one vocabulary, three compilers, and nobody types `integrated_multimodal_description:`. Replacing the prompt field entirely — a model reading intent instead of a table matching pills — is the same idea with a better front half.

Fragments are the expected input, not the degraded case. People think in bursts: a correction, a second thought that contradicts the first, the real point arriving third. AI writing is caught precisely because it is too smooth, and a tool that demands one clean flowing paragraph is demanding that a human produce the artefact of a machine before the machine will listen. Out of order, incomplete, self-correcting, arriving in pieces over a minute — that is the normal shape of intent, and the system’s job is to take it that way.

Intent is the durable record; the prompt is a receipt. The sidecar already keeps what you typed and what you chose alongside the compiled string, and the gallery prefers the typed one. That reads as a legibility choice and is really the deeper one: it pays off the moment a model is replaced, because intent recompiles for whatever comes next while a stored prompt is worth nothing to a checkpoint that wants a different grammar.

And a standing veto that will look unhelpful in a pull request. Nothing in this application browses, searches or recommends somebody else’s weights. Pulling a specific file you were sent is fine — that is what the Drive route is for. A marketplace is not, because a “discover models” surface is exactly the feature that would look obviously helpful and would undo every argument above it.

THE RISK WORTH NAMING

It is not the models, the latency or the scope. It is month four, when something does not fit cleanly and the cheapest available fix is a panel. That is the moment this becomes a node graph with better typography — and every rule in the section above exists to be the thing that stops it.